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Signal Approximations Based on Nonlinear and Optimal Piecewise Affine Functions

El Hadji S. Diop[1] · Ata Ngom[1] · V. B. Surya Prasath[2,3,4,5]

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Abstract

In this work, we address the problem of piecewise affine approximations, that is, to find piecewise affine functions that well-approximate a given signal. The proposed approach is optimal in the sense of L^2 norm and formulated in a compact and explicit way; no fitting stage is needed. Also, affine parameters are obtained as closed formulas, and affine approximation functions are optimal in their corresponding subdomains. In addition, we state and prove a recursive formula for approximation errors, which makes the approach optimal and nonlinear, links also the subdomains and helps derive an algorithm of complexity of order $O(MN^2)$, where M represents the number of piecewise affine approximants and N is the number of samples of the processed signal. Finally, obtained qualitative and quantitative results show that the presented method obtains good approximations and provides improvement over piecewise constant approximations.

Keywords Nonlinearity · Optimization · Approximations · Piecewise affine functions

✉ V. B. Surya Prasath
surya.iit@gmail.com

El Hadji S. Diop
ehsdiop@hotmail.com

Ata Ngom
atangom93@gmail.com

1 NAGIP-Nonlinear Analysis and Geometric Information Processing Group, Department of Mathematics, University Iba Der Thiam of Thies, BP 967, Thies, Senegal

2 Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center, Cincinnati 45229, OH, USA

3 Department of Pediatrics, University of Cincinnati College of Medicine, Cincinnati 45257, OH, USA

4 Department of Biomedical Informatics, University of Cincinnati, Cincinnati 45267, OH, USA

5 Department of Electrical Engineering and Computer Science, University of Cincinnati, Cincinnati 45221, OH, USA

1 Introduction

Dealing with real-world problems implies providing solutions that are computationally feasible. The need of rapid data processing constrained by low (limited) resources make new theoretical and practical approximation methods very demanding, maintaining the research field of approximations very active. Many applications of approximations can be found; for instance, in hybrid systems [4, 9], image compression, audio coding, learning theory, filters/transforms, and compressed sensing [12, 13, 19, 21, 26, 31–34, 45]. The main goal of approximations is to solve the problem of a likely complicated function (so called the target function) by using even simpler and easier computing functions (so called the approximants). As a consequence, the central question about the resolution and approximation complexity trade-off naturally raised [16]. In other words, this means that theoretical tools have to be combined with numerical analysis to provide new breakthrough in that domain. Interested readers could have a look at [17] for a good background on theoretical nonlinear approximations and some applications. We note that classically the spline functions [7, 10, 15, 30, 51, 52] have been considered by researchers, among the most illustrious approximation techniques in the signal processing area. Splines are very suitable and convenient, also, optimal rigorous estimations are given from limited data. We also note the primary approximations theoretical work [42], where various approaches were presented; for instance, polynomial functions, best approximations from linear space with regard to the minimax, least squares and L^1 norms.

In approximation theory, the basis of an exact signal approximation is built thanks to a linear combination of a relatively small number of vectors selected from that basis. This key concept makes orthogonal bases very essential in the field of signal processing, because signals can then be well-approximated with just a few components. There exists a very large family of approximation approaches. Many of them are based on harmonic analysis tools, for instance, Fourier analysis [6], wavelets [35, 37, 41] or chirplets [44]. Approximations can also be performed by superposing sigmoidal functions [14], using data-adaptive normalized Gaussian functions [43] or a gradient descent 'boosting' paradigm [24]. We also cite the recent sparse approximations [36], many solvers have been proposed [21, 27, 46, 49, 55]. In this study, we put more emphasis on piecewise smooth approximations, namely piecewise constant and affine approximations, which have found important applications in signal/image processing and related fields. For instance, we refer to the well-known Mumford-Shah model and extensions [39] for 2D. In a different paradigm, piecewise constant Gaussians were used in a Bayesian inference method [38] or stochastic models [50]. A piecewise-constant and low-rank approximations have also been proposed [20, 54, 56]. We finally refer the works of [5, 23, 47, 53] where piecewise affine approximations were combined to solve an induced optimization problem and different addressed issues.

1.1 Motivations

In this present work, we essentially focus on piecewise affine approximations. Our main motivation is the fact that there are many cases where piecewise constant func-

tions cannot well-approximate target functions. This is the case for bounded variations functions (BV[1]), see [2, 22, 25] for more details on BV spaces. The singularities of BV functions are the principal causes of such difficulties. This was overcome by proposing piecewise affine approximants on uniform regular triangulations [29]. A two-stage piecewise affine approximations method was proposed [48], approximations being gained from the data. In the so-called Lebesgue piecewise affine approximation work [3], the state spaces were partitioned on the basis of the vector fields variation. In other cited existing works, the mathematical framework is very well presented; however, authors did not provide well-established algorithms facilitating their implementations. In our work, in addition to a theoretical study, an algorithm is provided which makes the implementation of the approach possible and easy. We choose to show the superiority of the piecewise affine approach in comparison with the piecewise constant method.

1.2 Main Contributions

The proposed approximation method is nonlinear. Notice that contrary to linear approximations where approximants belong to linear spaces, approximants in nonlinear approximations come from nonlinear manifolds. For instance, comparisons of approximation rate between linear and nonlinear approximations were performed [16], and had showed the improvements of nonlinear approximations over linear ones. Our approach is close to the work of Szúcs et al [48] however with major differences. We summarize the main contributions as follows:

- A compact and explicit formulation of the affine approximation problem, contrary to [48] where a two-stage method was proposed. There is no need for a fitting stage.
- Proposition of a nonlinear and optimal piecewise affine approximation method. Affine parameters are obtained as closed formulas, and affine approximants are optimal in subdomains.
- Statement and proof of a recursive formula for approximation errors, making our approach optimal, linking also the subdomains and helping derive the algorithm.
- Proposition of approximation algorithm with a complexity of order $O(MN^2)$, M being the number of piecewise affine approximants and N the number of samples of the processed signal.

[1] Let $u \in L^1(\Omega)$, and denote

$$\int_\Omega |Du| = \sup\left\{\int_\Omega u \operatorname{div}(\varphi)\mathrm{d}x;\ \varphi = (\varphi_1, \varphi_2) \in \mathbb{C}_0^1(\Omega, \mathbb{R}^2) \text{ and } \|\varphi\| \le 1\right\},$$

where $\mathbb{C}_0^1(\Omega)$ is the space of continuously differential functions with compact support in Ω, $\operatorname{div}(\varphi) = \frac{\partial \varphi_1}{\partial x_1} + \frac{\partial \varphi_2}{\partial x_2}$ where derivatives are taken in a distributional sense, and $\|\varphi\| = \|(\varphi_1^2 + \varphi_2^2)^{1/2}\|_{L^\infty(\Omega)}$. The space of functions of bounded variation on Ω, denoted $\mathrm{BV}(\Omega)$, is the set of functions defined by

$$BV(\Omega) = \left\{u \in L^1(\Omega);\ \text{s.t.} \int_\Omega |Du| < \infty\right\}.$$

- Illustration of the good behavior of the proposed approximation approach. Qualitative and quantitative evaluations are provided.

1.3 Manuscript Organization

The rest of the paper is organized as follows. We recall some approximation methods in Sect. 2. Sections 3 and 4 are dedicated to the proposed approach, where the derived algorithm and obtained experimental results are presented, respectively. The paper ends in Sect. 5 with concluding remarks and some perspectives.

2 Review on Approximation Methods

2.1 Piecewise Affine Approximations

Let $\emptyset \neq E \subset \mathbb{R}$, $f : E \longrightarrow \mathbb{R}$ a one-dimensional nonlinear function to be approximated by a piecewise affine function. The problem is formulated as [48]:

$$\begin{aligned} &\min_{a_i, c_i, r_i} \sum_{i=1}^{N} \left(\int_{r_{i-1}}^{r_i} (f(z) - (a_i z + c_i))^2 \mathrm{d}z \right) \text{ s.t.} \\ &\underline{z} \leq r_1 \leq \cdots \leq r_{N-1} \leq \overline{z} \text{ and } a_i r_i + c_i = a_{i+1} r_i + c_{i+1}, \end{aligned} \tag{1}$$

where $r_0 = \underline{z}$, $r_N = \overline{z}$, E being assumed to be a closed set, then, $E = [\underline{z};\ \overline{z}]$, $(R_i)_{i=1}^N$ being a non-overlapping partition of E into N parts. We note that the problem (1) could be expressed as a nonlinear optimization problem, which could be solved for finding local optimizer or global optimization techniques [1, 11, 40]. Piecewise affine approximations were also derived for multivariate separable and nonseparable functions. f should be explicitly known or gained from a fitting process, which constituted the first step of the method. In fact, given the input data $(z_i;\ i = 1, \ldots, T)$, the fitting strategy consisted in representing all pairs (z_i, y_i) within a nonlinear function f s.t. $y = f(z)$, and:

$$f(z) = \sum \alpha_i f_i(z), \tag{2}$$

with $f_i : \mathbb{R}^n \longrightarrow \mathbb{R}$ being pre-defined basis functions and $\alpha_i \in \mathbb{R}$ are scalar parameters. The fitting procedure was then formulated as to minimize the error:

$$e_{fit} = \sum_{i=1}^{T} (y_i - f(z_i))^2, \tag{3}$$

where coefficients α_i of f in (2) have to be determined. Authors proposed a basis of powers $f_i(z) = z^{i-1}$, $i = 1, \ldots, n$ or polynomials:

$$f_i(z) = \sum_{i_1 + \cdots + i_n \leq n} z_1^{i_1} \cdots z_n^{i_n}.$$

They also suggested a direct minimization for solving (3), alternatives could be to heuristically solve the 1-norm of $\alpha = (\alpha_i)_{i=1}^n$ [8]:

$$\min \sum_{i=1}^{T} (y_i - f(z_i))^2 + \gamma \|\alpha\|_1. \tag{4}$$

It could also be solved using [4]:

$$\begin{aligned} &\min \sum_{i=1}^{T} (y_i - f(z_i))^2 + \gamma \sum_{i=1}^{n} \delta_i \text{ s.t.} \\ &- M\delta_j \leq \alpha_j \leq M\delta_j; \;\; j = 1, \ldots, n; \end{aligned} \tag{5}$$

with M being a sufficiently large number, $\delta_j = 0$ or 1 s.t. $(\alpha_j = 0) \Longrightarrow (\delta_j = 1)$.

2.1.1 Piecewise Affine Approximations in BV

We previously said that BV functions cannot be well-approximated with piecewise constant functions. In fact, one has:

Proposition 1 ([29]) *Let $m \in \mathbb{N}$, $u \in C_c^\infty(B(0,1), \mathbb{R}^m)$ a not identically zero function. Then, there is no sequence of piecewise constant functions $u_j \in BV(B(0,1), \mathbb{R}^m)$ s.t. $u_j \longrightarrow u$ $< \cdot >$-strictly in BV.*

The following gives a piecewise affine approximation result:

Theorem 2 ([29]) *Let $m \in \mathbb{N}$, $\Omega \subset \mathbb{R}^d$ lipschitz and bounded, λ a finite Borel measure on Ω and $u \in BV(\Omega, \mathbb{R}^m)$. Then, $\forall\ \varepsilon > 0$, there exists a countable family $\mathcal{R}$ of (rotated) rectangles and simplices $R \subset \Omega$ and $v \in W^{1,1}(\Omega, \mathbb{R}^m)$[2] satisfying:*

(i) *Sets in $\mathcal{R}$ are non-overlapping and $(\mathcal{L}^d + |Du|)(\Omega \setminus \cup \mathcal{R}) = 0$.*
(ii) *$\forall\ R \in \mathcal{R}$, the restriction $v|_R$ is affine.*
(iii) *$\|u - v\|_{L^1(\Omega, \mathbb{R}^m)} + |(Du)(\Omega) - (Dv)(\Omega)| < \varepsilon$.*
(iv) *$\mathcal{R} = \mathcal{R}_g \cup \mathcal{R}_b$, where $\lambda(\cup \mathcal{R}_b) < \varepsilon$ and*

$$\sum_{R \in \mathcal{R}_g} \int_{\partial R} |u - v| \mathrm{d}\mathcal{H}^{d-1} < \varepsilon.$$

(v) *$v|_{\partial\Omega} = u|_{\partial\Omega}$.*

[2] $W^{1,1}(\Omega, \mathbb{R}^m)$ is the Sobolev space defined by all functions $u \in L^1(\Omega, \mathbb{R}^m)$ such that there exists $g_1, \ldots, g_d \in L^1(\Omega, \mathbb{R}^m)$ verifying:

$$\int_\Omega u \frac{\partial \varphi}{\partial x_i} = -\int_\Omega g_i \varphi, \;\; \forall \varphi \in C_c^\infty(\Omega) \text{ and } \forall i = 1, \ldots, d$$

Let us provide some rough explanation of Theorem 2. Basically, the non-overlapping family $\mathcal{R}$ should be such that it recovers the domain Ω, meaning that the measure (thanks to Radon–Nikodym decomposition) of $\Omega - \cup R = 0$. Also, the restriction of the approximation v on $\mathcal{R}$ should be affine. The Sobolev space $W^{1,1}$ is used in Theorem 2, while the L^2 is used in our study; the approximation is thus done with $W^{1,1}$ norm. The approximation v and $u \in BV(\Omega)$ should coincide in the boundary of its domain $\partial\Omega$ (the need of its regularity makes the choice Ω to be Lipschitz).

2.2 Piecewise Constant Approximations

2.2.1 Linear Approximations

Consider $N \in \mathbb{N}^\star$ and an ordered set T of points $0 = t_0 < t_1 < \cdots < t_N = 1$ in $\Omega = [0,\ 1]$. Such subdivision constitutes a partition $\Pi = \{I_k\}_{k=1}^N$ of Ω into N disjoint intervals $I_k = [t_{k-1};\ t_k]$, $1 \le k \le N$. $\mathcal{S}_1(T)$ represents the space of piecewise constant functions with regard to the partition, the family $\{\chi_I : I \in \Pi\}$ of characteristic functions constitutes a basis of $\mathcal{S}_1(T)$. This means that $\mathcal{S}_1(T)$ is a linear space, its dimension is equal to N.

The following approximation error of $h \in L^p(\Omega)$ ($0 < p \le \infty$) was introduced [16]:

$$s(h, T)_p = \inf_{S \in \mathcal{S}_1(T)} \|h - S\|_{L^p(\Omega)}. \tag{6}$$

Let $0 < \alpha \le 1$, $0 < p \le \infty$ and $K > 0$. Lip$(\alpha, L^p(\Omega))$ denotes the set of functions $h \in L^p(\Omega)$ such that $\|h(\cdot + h) - h\|_{L^p([0;\ 1-h])} \le K\delta^\alpha$, $0 < \delta < 1$. The space is endowed with the following norm:

$$|h|_{Lip(\alpha, L^p(\Omega))} = \max_{t>0} t^{-\alpha} \left\{ \max_{|\delta| \le t} \|\sum_{k=0}^{r} (-1)^{r-k} \binom{r}{k} h(\cdot + k\delta)\|_{L^p} \right\}$$

.

The quantity $S_T \in \mathcal{S}^1(T)$ was defined as:

$$S_T(x) = \frac{1}{|I|} \int_I h \mathrm{d}t,\ x \in I,\ I \in \Pi, \tag{7}$$

with $|I|$ representing the Lebesgue measure of I. It followed then [18]:

$$s(h, T)_p \le \|h - S_T\|_{L^p(\Omega)} \le C_p |h|_{Lip(\alpha, L^p(\Omega))} \delta_T^\alpha, \tag{8}$$

C_p being a constant depending at most on p, $\delta_T = \max_{0 \le k < N} |t_{k+1} - t_k|$.

Denoted $\triangle_n = \{\frac{k}{n};\ 0 \le k \le n\}$ the set of n equally spaced points from Ω, the following result was provided:

Proposition 3 ([16]) *Under assumptions of h satisfying $s(h, \triangle_n)_p \le Kn^{-\alpha}$, $n = 1, 2, \ldots$, for some $0 < \alpha \le 1$ and $K > 0$, one has: $h \in Lip(\alpha, L^p(\Omega))$ and $|h|_{Lip(\alpha, L^p(\Omega))} \le C_p K$.*

2.2.2 Nonlinear Approximations

Denote T the finite set of points $0 = t_0 < t_1 < \cdots < t_n = 1$ from $\Omega = [0,\ 1]$. Let $\mathcal{S}^1(T)$ be the set of piecewise constant functions S with breakpoints from T. Consider the nonlinear space $\sum_n = \bigcup_{\#T=n+1} \mathcal{S}^1(T)$, $\#T$ being the cardinality of T. Any function belonging to $\sum_n$ is piecewise constant with at most n pieces.

Let $h \in L^p(\Omega)$ $(0 < p \leq \infty)$, the following L^p nonlinear approximation error $\sigma_n(h)_p$ of h was introduced [16]:

$$\sigma_n(h)_p = \inf_{S \in \sum_n} \|h - S\|_{L^p(\Omega)}. \tag{9}$$

Proposition 4 ([16, 28]) *Let $h \in C(\Omega)$. Thus,*

$$\sigma_n(h)_\infty \leq \frac{K}{2n},\ n = 1, 2, \ldots,\ \textit{if and only if } h \in BV(\Omega). \tag{10}$$

3 Optimal Nonlinear Piecewise Affine Approximation

Let $N \in \mathbb{N}^\star$, $p \in [[0;\ N]]$, $k \in [[0;\ N]]$ and $l \in [[0;\ k-1]]$. We denote $[[1;\ N]] := \{1, 2, \ldots, N\}$. For any discrete signal x, $x[n]$ denotes its samples, $x_p^k[n]$ represents the piecewise affine signal defined on $[[0;\ k]]$, τ_p^k denotes the l^2 norm on $[[0; k]]$ and δ_l^k is defined as $\delta_l^k = \sum_{n=l}^{k} |x[n] - x_p^k[n]|^2$. Let us consider an N-sample real-valued signal $x : [[0;\ N]] \longrightarrow \mathbb{R}; n \mapsto x[n]$.

Let $p \in [[0;\ N]]$ and $k \in [[0;\ N]]$ be fixed values. Then, we denote by $x_p^k[n]$ the piecewise affine signal defined on $[[0;\ k]]$ as follows:

$$x_p^k[n] = \alpha_p^k n + \beta_p^k, \tag{11}$$

α_p^k, β_p^k are constant values to be determined. We assume $x_p^k[n]$ takes at most p different values and minimizes the l^2 square norm as follows:

$$\tau_p^k = \|x - x_p^k\|_{[[0;k]]}^2 := \sum_{n=0}^{k} |x[n] - \alpha_p^k[n] - \beta_p^k|^2. \tag{12}$$

Equation (12) measures the difference in l^2 sense between the signal $x[n]$ and its piecewise affine approximations. Therefore, the proposed approach will be optimal since all affine parameters α_p^k and β_p^k in every subdomain are obtained by minimizing an error (12), and the optimality is in L^2 sense.

Next is a result on how to determine the affine parameters:

Proposition 5 *Let $k \in [[0;\ N]]$, $l \in [[0;\ k-1]]$, and define δ_l^k as following:*

$$\delta_l^k = \sum_{n=l}^{k} |x[n] - \alpha_p^k n - \beta_p^k|^2. \tag{13}$$

Then, the affine parameters α_l^k and β_l^k that minimize δ_l^k are given as follows:

$$\alpha_l^k = \frac{\sum_{n=l}^{k}(x[n]-\bar{x})(n-\bar{n})}{\sum_{n=l}^{k}(n-\bar{n})^2}, \text{ and} \tag{14}$$

$$\beta_l^k = \bar{x} - \frac{\sum_{n=l}^{k}(x[n]-\bar{x})(n-\bar{n})}{\sum_{n=l}^{k}(n-\bar{n})^2}\bar{n}, \tag{15}$$

with $\bar{n} = \frac{1}{1+k-l}\sum_{n=l}^{k} n$ and $\bar{x} = \frac{1}{1+k-l}\sum_{n=l}^{k} x[n]$.

Proof Let us consider the differentiable mapping $f : \mathbb{R} \longrightarrow \mathbb{R}_+;\ f(\alpha,\beta) = \sum_{n=l}^{k} |x[n] - \alpha n - \beta|^2$. f being convex, then, any local minimum is a global minimum. Thus, the affine parameters α_l^k and β_l^k minimizing f satisfy $\frac{\partial f}{\partial \alpha}(\alpha_l^k, \beta_l^k) = \frac{\partial f}{\partial \beta}(\alpha_l^k, \beta_l^k) = 0$. It follows then:

$$\sum_{n=l}^{k} n(x[n] - \alpha_l^k n - \beta_l^k) = 0, \text{ and} \tag{16}$$

$$\sum_{n=l}^{k} (x[n] - \alpha_l^k n - \beta_l^k) = 0. \tag{17}$$

From (17), one gets:

$$\beta_l^k = \bar{x} - \alpha_l^k \bar{n} = 0, \tag{18}$$

and expanding (16) yields:

$$\sum_{n=l}^{k} n x[n] - \alpha_l^k \sum_{n=l}^{k} n^2 - \beta_l^k \sum_{n=l}^{k} n = 0. \tag{19}$$

Finally, one concludes by combining (18) in (19). □

The next result is a recursive formula for determining the approximations errors τ_p^k (12):

Proposition 6 *For all p and $k \in [[0;\ N]]$, one has:*

$$\tau_p^k = \min_{l \in [[0;\ k-1]]} [\tau_{p-1}^l + \delta_{l+1}^k]. \tag{20}$$

Proof : Let us evaluate the terms into brackets. For all $l \in [[0;\ k-1]]$, one has:

$$\tau_{p-1}^l + \delta_{l+1}^k = \sum_{n=0}^{l} |x[n] - \alpha_{p-1}^l n - \beta_{p-1}^l|^2 + \sum_{n=l+1}^{k} |x[n] - \alpha_{l+1}^k n - \beta_{l+1}^k|^2. \tag{21}$$

Now, let h be the piecewise affine mapping $h : [[0;\ k]] \longrightarrow \mathbb{R}$ defined by:

$$h[n] = \begin{cases} \alpha_{p-1}^l n + \beta_{p-1}^l, & \text{if } n \in [[0;\ l]] \\ \alpha_{l+1}^k n + \beta_{l+1}^k, & \text{if } n \in [[l+1;\ k]] \end{cases}. \tag{22}$$

It follows then (21) can rewrite as follows:

$$\sum_{n=0}^{l} |x[n] - \alpha_{p-1}^l n - \beta_{p-1}^l|^2 + \sum_{n=l+1}^{k} |x[n] - \alpha_{l+1}^k n - \beta_{l+1}^k|^2 = \sum_{n=0}^{l} |x[n] - h[n]|^2 + \sum_{n=l+1}^{k} |x[n] - h[n]|^2, \tag{23}$$

if and only if $\tau_{p-1}^l + \delta_{l+1}^k = \sum_{n=0}^{k} |x[n] - h[n]|^2 = \|x - h\|_{[[0;\ k]]}^2$. Accounting that to x_p^k minimizes that norm, then, one gets:

$$\|x - h\|_{[[0;\ k]]}^2 \geq \|x - x_p^k\|_{[[0;\ k]]}^2 = \tau_p^k. \tag{24}$$

It follows then, one has:

$$\forall\, l \in [[0;\ k-1]],\ \tau_p^k \leq \tau_{p-1}^k + \delta_{l+1}^k. \tag{25}$$

The set where the signal $x[n]$ being discrete and finite, then one can say that τ_p^k is the maximal element of the finite family $\{\tau_{p-1}^k + \delta_{l+1}^k; l \in [[0;\ k-1]]\}$. Thus, one can conclude that:

$$\tau_p^k = \min_{l \in [[0;\ k-1]]} [\tau_{p-1}^l + \delta_{l+1}^k]. \tag{26}$$

□

4 Algorithm and Experimental Results

4.1 Piecewise Affine Approximation Algorithm

Thanks to relation (12), Eq. (20) in Proposition 6 can rewrite:

$$\sum_{n=0}^{k} |x[n] - x_p^k[n]|^2 = \min_{l \in [0;\ k-1]} \left[\sum_{n=0}^{l} |x[n] - x_{p-1}^l[n]|^2 + \sum_{n=l+1}^{k} |x[n] - \delta_{l+1}^k|^2 \right]. \tag{27}$$

We present below the details of the algorithm for obtaining the affine approximations x_M^N by computing in an iterative basis all x_p^k, for all $p \in [[1;\ M]]$ and for all $k \in [[0;\ N]]$, where $M, N \in \mathbb{N}^\star$ are fixed and $M \leq N$.

Thanks to the definition of x_p^k (11), one sees that x_0^k does not exist because it takes 0 value in $[[0;\ k]]$. Also, x_p^0 is defined only for $n = 0$, and using $\tau_{p,0}$ previously defined by equation (12), one has:

$$\forall\ p \in [[1;\ M]],\ x_p^0 = x[0]. \tag{28}$$

We recall that for all $k \in [[0;\ N]]$, $x_1^k[n]$ takes at most one affine term in $[[0;\ k]]$, which is given by (14) in Proposition 5:

$$\forall\ k \in [0;\ N],\ x_1^k[n] = \alpha_1^k n + \beta_1^k,\ n \in [[0;\ k]]; \tag{29}$$

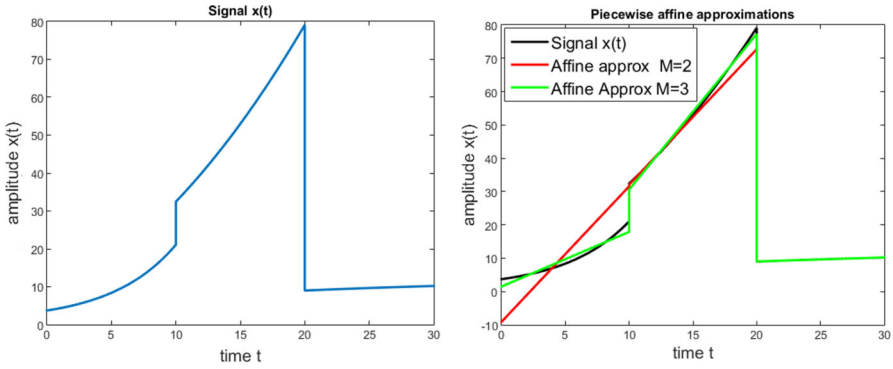
where α_1^k and β_1^k are respectively given by (14) and (15) with $\bar{n} = \frac{1}{k}\sum_{n=1}^{k} n$ and $\bar{x} = \frac{1}{k}\sum_{n=1}^{k} x[n]$.

Let us now consider the case $p = 2$. Denote l_2^k the index $l \in [[0;\ k-1]]$ realizing the minimum of the sum $\tau_1^l + \delta_{l+1}^k$, then, it follows:

$$\begin{aligned}
\sum_{n=0}^{k} |x[n] - x_2^k[n]|^2 &= \sum_{n=0}^{l_2^k} |x[n] - x_2^k[n]|^2 + \sum_{n=l_2^k+1}^{k} |x[n] - x_2^k[n]|^2 \\
&= \min_{l \in [[0;\ k-1]]} [\tau_1^l + \delta_{l+1}^k] \\
&= \sum_{n=0}^{l_2^k} |x[n] - \alpha_1^{l_2^k} n - \beta_1^{l_2^k}|^2 + \sum_{n=l_2^k+1}^{k} |x[n] - \alpha_{l_2^k+1}^k n - \beta_{l_2^k+1}^k|^2.
\end{aligned}$$

Thus, $x_2^k[n]$ is obtained by identification in the following:

$$x_2^k[n] = \begin{cases} \alpha_1^{l_2^k} n + \beta_1^{l_2^k}, & \text{if } n \in [[0;\ l_2^k]] \\ \alpha_{l_2^k+1}^k n + \beta_{l_2^k+1}^k, & \text{if } n \in [[l_2^k + 1;\ k]] \end{cases}, \tag{30}$$



(a) **(b)**

Fig. 1 **a** Signal $x(t)$ (32). **b** Piecewise affine approximations for $M = 2$ and $M = 3$ (Color figure online)

where α_i^j and β_i^j terms are given by (14) and (15) in Proposition 5.

In a similar way, affine approximations terms for the case $p = 3$ are given as follows:

$$x_3^k[n] = \begin{cases} x_2^{l_3^k}[n], & \text{if } n \in [[0;\ l_3^k]] \\ \alpha_{l_3^k+1}^k n + \beta_{l_3^k+1}^k, & \text{if } n \in [[l_3^k + 1;\ k]] \end{cases}, \tag{31}$$

where $x_2^k[n]$ is obtained thanks to (30) with $k = l_3^k$, α_i^j and β_i^j terms are given by (14) and (15) in Proposition 5.

Below is an example illustrating the affine approximations approach:

Example 1 Let $x(t)$ be the following continuous signal:

$$x : [0;\ 30] \mapsto \mathbb{R};\ t \longrightarrow x(t) = \begin{cases} 1 + \exp\left(1 + \dfrac{t}{5}\right), & \text{if } t \in [0;\ 10[\\ 4 + \left(2 + \dfrac{t}{3}\right)^2, & \text{if } t \in [10;\ 20[\\ \log(1 + t^3)\ , \text{if } t \in [20;\ 30]. \end{cases} \tag{32}$$

The signal $x(t)$ (32) and the respective piecewise affine approximations with $M = 2$ and $M = 3$ are displayed in Fig. 1. Notice for $M = 2$, the red line corresponds to the affine approximations of $x(t)$ in $[0;\ 20]$, while the signal and its approximations (red line) almost coincide on $[20;\ 30]$. For $M = 3$, the affine approximations are plotted in green. This example shows clearly the good behavior of the proposed approach.

In order to obtain x_M^N, the generalization follows by extending Eq. (31) for all $p \in [2;\ M]$ and for all $k \in [0;\ N]$ in the following:

$$x_p^k[n] = \begin{cases} x_{p-1}^{l_p^k}[n], & \text{if } n \in [[0;\ l_p^k]] \\ \alpha_{l_p^k+1}^k n + \beta_{l_p^k+1}^k, & \text{if } n \in [[l_p^{k+1};\ k]] \end{cases}, \tag{33}$$

$l_p^k \in [0;\ k-1]$ being the optimum minimizing $\tau_{p-1}^l + \delta_{l+1}^k$, α_i^j and β_i^j terms are given by (14) and (15) in Proposition 5. In other words, based on an recursive procedure, Eq. (33) looks for the signal $x[n]$ subdomains on which it best (l^2 sense) fits to its affine approximations in such subdomains.

Let $x[n]$ be the processing signal, M a user given parameter and N the number of samples of $x[n]$. Using the proposed approach, one is looking for an optimal piecewise affine approximation of the signal, denoted x_M^N. The complexity is of order O(MN^2); this means that MN^2 operations are needed in order to obtain the piecewise affine approximate x_M^N. The algorithm details can be found in Algorithm 1.

As in approximation technique, $M << N$ should be smaller than N, but, of course the approximation error $\tau_l^k \rightarrow 0$ as $M \rightarrow N$. For instance, given an N-sample piecewise affine $x[n]$ signal, its reconstruction is perfect (0 error) for $M \sim N$.

Remark 1 In fact, M fixes the number of piecewise affine approximants on the M subdomains $\{S_m;\ m \in [[1;\ M]]\}$. In every S_m, the corresponding piecewise affine approximant of $x[n]$ on S_m is determined iteratively thanks to formula (33). Thus, the Algorithm 1 converges.

Algorithm 1 Piecewise affine approximation algorithm.

Input: Signal $x[n]$, M and N.
Output: Approximation x_M^N.

```
for k = 1 to N do
    for l = 1 to k do
        Compute α_l^k (14), β_l^k (15) and δ_l^k (13)
    end for
end for
for p = 1 to M do
    Compute x_p^0 (28)
end for
for k = 0 to N do
    Compute x_1^k (29)
end for
for p = 2 to M do
    for k = 1 to N do
        Find l* = arg min_{l∈[[0; k−1]]} [τ_{p−1}^l + δ_{l+1}^k]
    end for
    for k = 2 to N do
        Compute x_p^k (33)
    end for
end for
```

4.2 Experimental Results

This section is dedicated to the analysis of the results obtained with Algorithm 1. The efficiency of the proposed approximations approach is illustrated on various signals, and obtained results are compared qualitatively and quantitatively to [20]. For

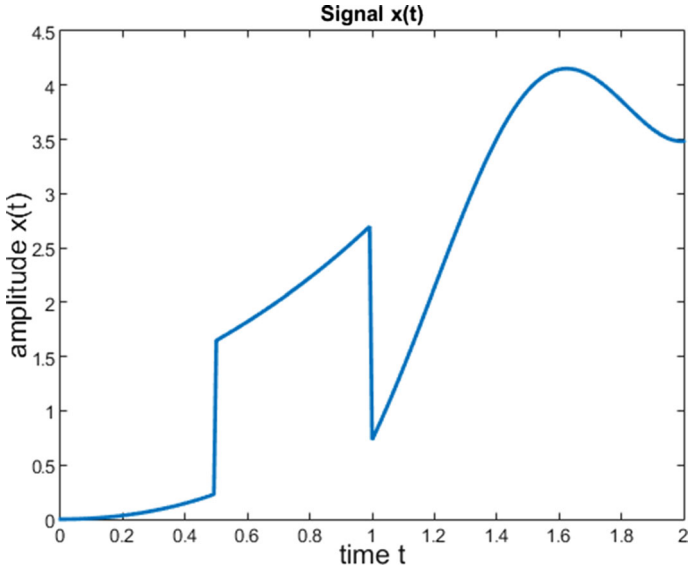


Fig. 2 Signal $x_1(t)$ (34)

the quantitative results, approximations errors are computed with L^2 norm and the minimum squared error (MSE).

Let us consider first the following non-stationary signal $x_1 : [0;\ 2] \mapsto \mathbb{R}$ defined $\forall\, t \in [0;\ 2]$ by:

$$x_1(t) = \begin{cases} t^2 \sin(1+t^2), & \text{if } t \in [0;\ 1[\\ \exp(t)\ , \text{if } t \in [1;\ \frac{3}{2}[& \\ 2\log(1+t^3) + \cos((1+t)^2), & \text{if } t \in [\frac{3}{2};\ 2] \end{cases}. \tag{34}$$

We represent the signal $x_1(t)$ (34) in Fig. 2. $x_1(t)$ has 257 samples, approximations are performed for $M = 20, 40$, and 90. This means that $x_1(t)$ is approximated with M affine functions. The different approximation results are displayed in Fig. 3a, b, showing the good behavior of the proposed method. Indeed, one sees in Fig. 2a that the signal $x_1(t)$ almost coincides with the approximations obtained for $M = 40$ (in blue) and $M = 90$ (green) piecewise affine functions. This is confirmed with the quantitative results reported in Table 1. With $M = 20$, the piecewise affine approach behaves in a good way in [0; 1.4[, but results are worse (red) than for $M = 40$ and $M = 90$ in [1.4; 2]. This can be expected, the higher the number of piecewise affine functions, the better the approximation results. The comparison with piecewise constant functions (Fig. 2b) reveals the improvements of the piecewise affine method.

The approach is also illustrated on the following non-stationary signal $x_2 : [0;\ 3] \mapsto \mathbb{R}$ defined $\forall\, t \in [0;\ 3]$ by:

$$x_2(t) = 1 + \frac{x^4}{2} + (5\sqrt{x} - 4)\sin(x^2 + 7x - 5) + \left(\frac{x}{2} + 7\right)\cos(2\pi\sqrt{3x+2}). \tag{35}$$

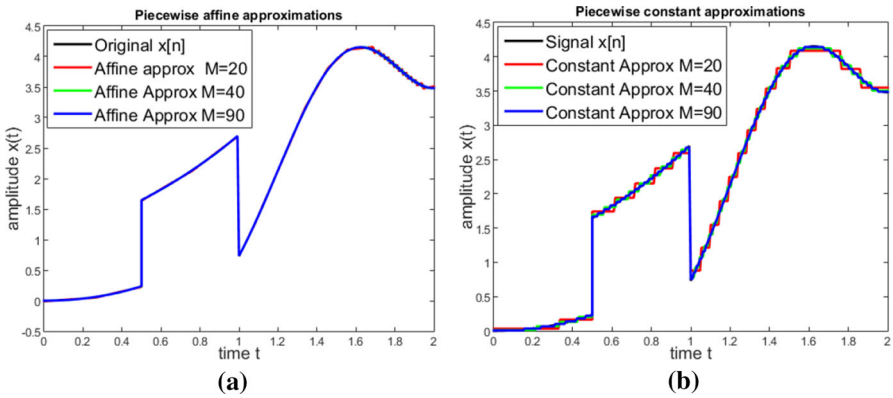


Fig. 3 Piecewise approximations of $x_1(t)$ (34) with $M = 20$ (red), $M = 40$ (green) and $M = 90$ (blue). **a** Affine. **b** Constant (Color figure online)

Table 1 Approximation errors between $x_1(t)$, $x_2(t)$ and $x_n(t)$, and corresponding approximations obtained with the approach and [20]

		L^2 norm	MSE
Signal $x_1(t)$ (34): $M = 20$	Proposed	**0.22**	$\mathbf{(1.9)10^{-3}}$
	Method [20]	1.06	$(4.4)10^{-3}$
$M = 40$	Proposed	**0.11**	$\mathbf{(4.7)10^{-5}}$
	Method [20]	0.52	$(1.07)10^{-3}$
$M = 90$	Proposed	0.1	$(4.5)10^{-5}$
	Method [20]	0.22	$(1.9)10^{-4}$
Signal $x_2(t)$ (35): $M = 20$	Proposed	**9.01**	**0.21**
	Method [20]	18.21	0.86
$M = 40$	Proposed	**4.16**	**0.04**
	Method [20]	9.7	0.24
$M = 90$	Proposed	**2.01**	**0.01**
	Method [20]	4.38	0.05
Signal $x_n(t)$: $K = 120$	Proposed	20.05	1.04
	Method [20]	19.35	0.97

The signal $x_2(t)$ (35) has 385 samples, its representation is showed in Fig. 4. Approximation results with $M = 20$, 40 and 90 can be found in Fig. 5a, b, showing again the good results of the affine based approximation approach.

However, the proposed approach is not robust in a noisy environment. For instance, we illustrate this by adding a Gaussian noise of about an signal-to-noise ratio (SNR) per sample equal to 10^{-3} db to the following signal:

$$x : [0;\ 1] \mapsto \mathbb{R};\ t \longrightarrow x(t) = (5\sqrt{t} - 4)\sin(t^2 + 7t - 5) + \left(\frac{t}{2} + 7\right)\cos(2\pi\sqrt{30t + 8}). \tag{36}$$

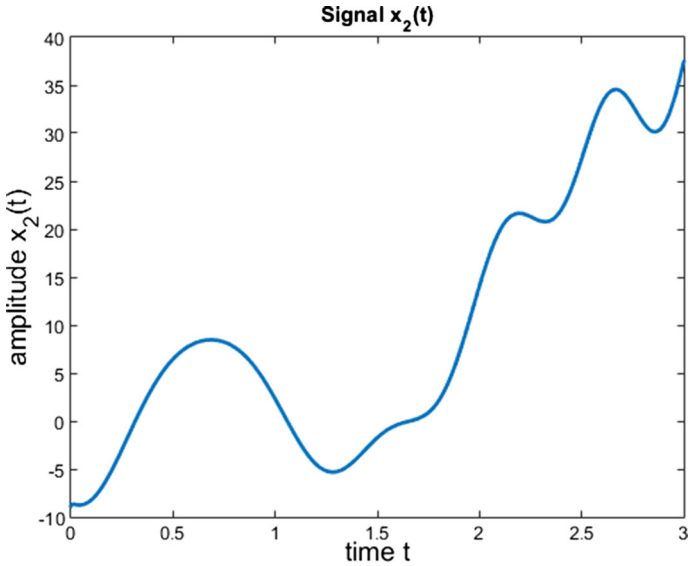


Fig. 4 Signal $x_2(t)$ (35)

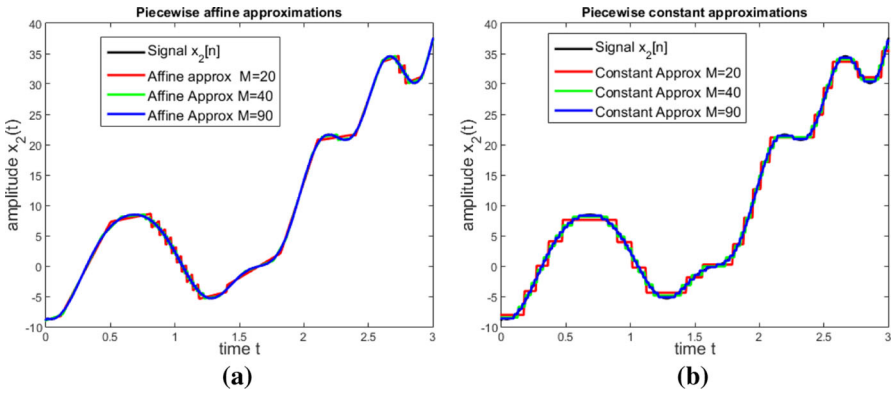


Fig. 5 Piecewise approximations of $x_2(t)$ (35) with $M = 20$ (red), $M = 40$ (green) and $M = 90$ (blue). **a** Affine. **b** Constant (Color figure online)

The resulting signal is denoted $x_n(t)$, its plotting and approximations are respectively displayed in Figs. 6 and 7.

$x_(t)$ has 120 samples. Figure 7 clearly shows the lack of robustness against noise of both the proposed method and approach [20]. Indeed, even with $K = 120$, neither method cannot recover the uncorrupted signal $x_(t)$.

In addition to the provided qualitative results showing the effectiveness of the proposed approach, quantitative evaluations are also performed. To do so, we compute the L^2 norm and the MSE between the true signal values ($x_1(t)$, $x_2(t)$ and $x_n(t)$) and the approximations obtained with the proposed method and approach developed in

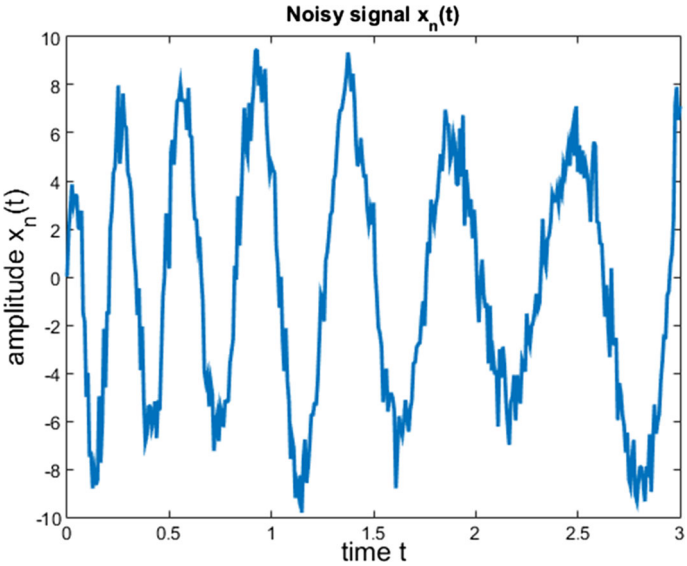
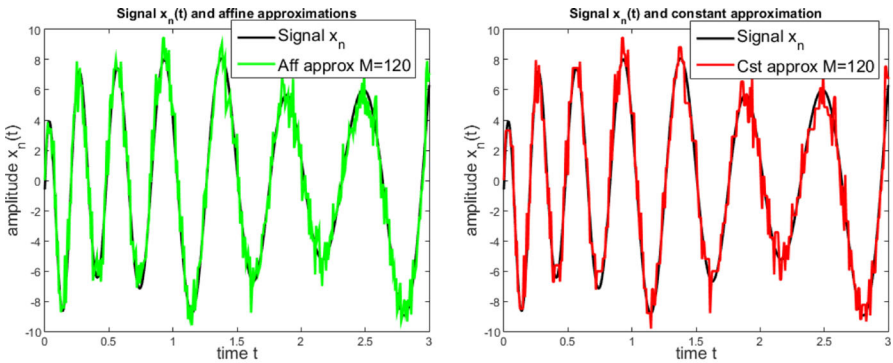


Fig. 6 Noisy signal $x_n(t)$



(a) **(b)**

Fig. 7 Piecewise approximations of $x_n(t)$ with $M = 120$. **a** Affine. **b** Constant (Color figure online)

[20]. Quantitative results are reported in Table 1 where best results are put in bold, showing that improvements of the proposed approach can go up to five times.

5 Conclusions

We have addressed in this work the problem of piecewise affine approximations which consists in finding piecewise affine functions that well-approximate a given signal. Besides the fact that the proposed approach is optimal in the sense of L^2 norm, we have formulated such problem in a compact and explicit way so that no fitting stage is needed as it the case in other methods. Also, affine parameters are obtained as closed

formulas, and affine approximation functions are optimal in their corresponding subdomains. Moreover, we have proposed a recursive formula for approximation errors, and the proof is provided. The latter formula has made the approach optimal, it also links the subdomains, and most of all, it helps derive the piecewise affine algorithm which is of complexity of order $O(MN^2)$, M representing the number of piecewise affine approximants and N being the number of samples of the processed signal. The complexity could be seen as a bit high; however, the number of parameters should be correlated to the computational effort, for instance, the nonlinearity, the optimality, closed formulas for affine parameters. Finally, obtained qualitative and quantitative results have showed that the proposed approach has a good behavior and improves a lot on piecewise constant approximations. As for future perspectives in this direction, it could be interesting to optimize also the number of piecewise affine approximants or equivalently the number of subdomains; however, this would increase the overall complexity, and one could face a memory issue especially for signals with a big or relatively big size. Moreover, making the approximation method robust in a noisy environment and combining the proposed approach with sparse regression are other potential directions.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

1. C. Adjiman, I. Androulakis, C. Maranas et al., A global optimization method, αBB, for process design. Comput. Chem. Eng. **20**, S419–S424 (1996)
2. L. Ambrosio, N. Fusco, D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems* (Oxford Mathematical Monographs, Oxford, 2000)
3. S. I. Azuma, J. I. Imura, T. Sugie, Lebesgue piecewise affine approximation of nonlinear systems. Nonlinear Anal. Hybrid Syst. **4**(1), 92–102 (2010)
4. A. Bemporad, M. Morari, Control of systems integrating logic, dynamics, and constraints. Automatica **35**(3), 407–427 (1999)
5. A. Bemporad, A. Oliveri, T. Poggi et al., Ultra-fast stabilizing model predictive control via canonical piecewise affine approximations. IEEE Trans. Autom. Control **56**(12), 2883–2897 (2011)
6. T. Blu, M. Unser, Quantitative Fourier analysis of approximation techniques. I. Interpolators and projectors. IEEE Trans. Signal Process. **47**(10), 2783–2795 (1999)
7. A. Bouchikhi, A. Boudraa, Multicomponent AM–FM signals analysis based on EMD–B-splines ESA. Signal Process. **92**(9), 2214–2228 (2012)
8. S. Boyd, L. Vandenberghe, *Convex Optimization* (Cambridge University Press, 2004)
9. M. Branicky, Studies in Hybrid Systems: Modeling, Analysis, and Control. PhD thesis, Massachusetts Institute of Technology, Massachusetts (1995)
10. I. Burova, A. Doronina, D. Zhilin, Splines of the fourth order approximation and the Volterra integral equations. WSEAS Trans. Math. **20**, 475–488 (2021)
11. B. Chachuat, A.B. Singer, P.I. Barton, Global methods for dynamic optimization and mixed-integer dynamic optimization. Ind. Eng. Chem. Res. **45**(25), 8373–8392 (2006)
12. D. Chen, S.B. Ko, A novel decimal logarithmic converter based on first-order polynomial approximation. Circuits Syst. Signal Process. **31**(3), 1179–1190 (2012)

13. R.J. Cintra, An integer approximation method for discrete sinusoidal transforms. Circuits Syst. Signal Process. **30**(6), 1481 (2011)
14. G. Cybenko, Approximation by superpositions of a sigmoidal functions. Math. Control Signals Syst. **2**, 303–314 (1989)
15. C. de Boor, *A Practical Guide to Splines* (Springer, New York, 1978)
16. R.A. DeVore, Nonlinear approximation. Acta Numer. **7**, 51–150 (1998)
17. R.A. DeVore, *Multiscale, Nonlinear and Adaptive Approximation* (Springer, Heidelberg, 2009), pp.169–201
18. R.A. DeVore, G.G. Lorentz, *Constructive Approximation, Chap 12* (Springer, Heidelberg, 1993), pp.354–389
19. S. Dhabu, A.P. Vinod, A new time-domain approach for the design of variable FIR filters using the spectral parameter approximation technique. Circuits Syst. Signal Process. **36**(5), 2154–2165 (2017)
20. E.H.S. Diop, A.O. Boudraa, V.B.S. Prasath, Optimal nonlinear signal approximations based on piecewise constant functions. Circuits Syst. Signal Process. **39**(5), 2673–2694 (2020)
21. D.L. Donoho, Compressed sensing. IEEE Trans. Inf. Theory **52**(4), 1289–1306 (2006)
22. L.C. Evans, R.F. Gariepy, *Measure Theory and Fine Properties of Functions. Studies in Advanced Mathematics* (CRC Press, Boca Raton, 1992)
23. E. Firippi, M. Chaves, Topology-induced dynamics in a network of synthetic oscillators with piecewise affine approximation. Chaos Interdiscip. J. Nonlinear Sci. **30**(11), 113–128 (2020)
24. J.H. Friedman, Greedy function approximation: a gradient boosting machine. Ann. Stat. **29**(5), 1189–1232 (2001)
25. E. Giusti, *Minimal Surfaces and Functions of Bounded Variation* (Birkhuser, Boston, 1994)
26. E.M. Hamed, A.M. AbdelAty, L.A. Said et al., Effect of different approximation techniques on fractional-order KHN filter design. Circuits Syst. Signal Process. **37**, 5222–5252 (2018)
27. Y. Isaac, Q. Barthélemy, C. Gouy-Pailler et al., Multi-dimensional signal approximation with sparse structured priors using split Bregman iterations. Signal Process. **130**, 389–402 (2017)
28. J.P. Kahane, *Teoria constructiva de functiones* (Course notes, University of Buenos Aires, 1961)
29. J. Kristensen, F. Rindler, Piecewise affine approximations for functions of bounded variation. Numer. Math. **132**(2), 329–346 (2016)
30. B.I. Kvasov, *Methods of Shape Preserving Spline Approximations* (World Scientific, Singapore, 2000)
31. F. Leulmi, Y. Ferdi, Improved digital rational approximation of the operator s^{α} using second-order s-to-z transform and signal modeling. Circuits Syst. Signal Process. **34**(6), 1869–1891 (2015)
32. Q. Li, B. Shen, Z. Wang et al., A sampled-data approach to distributed h ∞ resilient state estimation for a class of nonlinear time-delay systems over sensor networks. J. Frankl. Inst. **354**(15), 7139–7157 (2017)
33. Q. Li, B. Shen, Z. Wang et al., Anevent-triggered approach to distributed h∞ state estimation for state-saturated systems with randomly occurring mixed delays. J. Frankl. Inst. **355**(6), 3104–3121 (2018)
34. G. Makkena, M. Srinivas, Nonlinear sequence transformation-based continuous-time wavelet filter approximation. Circuits Syst. Signal Process. **37**(3), 965–983 (2018)
35. S. Mallat, *A Wavelet Tour of Signal Processing* (Academic Press, San Diego, 1999)
36. S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way* (Elsevier, San Diego, 2009)
37. S. Mallat, Z. Zhang, Matching pursuits with time-frequency dictionaries. IEEE Trans. Signal Process. **41**(12), 3397–3415 (1993)
38. R. Martin, W. Shen, Asymptotically optimal empirical bayes inference in a piecewise constant sequence model. Technical Report arXiv:1712.03848v1 (2017)
39. D.B. Mumford, J. Shah, Optimal approximations by piecewise smooth functions and associated variational problems. Commun. Pure Appl. Math. **42**(5), 577–685 (1989)
40. I. Papamichail, C. Adjiman, Global optimization of dynamic systems. Comput. Chem. Eng. **28**(3), 403–415 (2004)
41. Y.C. Pati, R. Razaiifar, P.S. Krishnaprasad, Orthogonal matching pursuit: recursive function approximation with applications to wavelet decomposition, in *27th Annual Asilomar Conference on Signals Systems and Computers* (IEEE, Pacific Grove, 1993), pp. 1–5
42. M.J.D. Powell, *Approximation Theory and Methods* (Cambridge University Press, 1981)
43. S. Qian, D. Chen, K. Chen, Signal approximation via data-adaptive normalized gaussian functions and its applications for speech processing, in *IEEE ICASSP* (San Francisco, 1992), pp. 141–144

44. S. Qian, D. Chen, Q. Yin, Adaptive chirplet based signal approximation, in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)* (Seattle, 1998), pp. 1781–1784
45. S.D. Roy, A new Chebyshev-like low-pass filter approximation. Circuits Syst. Signal Process. **29**(4), 629–636 (2010)
46. R. Rubinstein, M. Zibulevsky, M. Elad, Double sparsity: learning sparse dictionaries for sparse signal approximation. Trans. Signal Process. **58**(3), 1553–1564 (2010)
47. V. Sivaramakrishnan, A.P. Vinod, M.M.K. Oishi, *Convexified open-loop stochastic optimal control for linear non-gaussian systems*. Technical report, ArXiv (2020)
48. A. Szűcs, M. Kvasnica, M. Fikar, Optimal piecewise affine approximations of nonlinear functions obtained from measurements. IFAC Proc. Vol. **45**(9), 160–165 (2012)
49. R. Tibshirani, Regression shrinkage and selection via the lasso. J. R. Stat. Soc. Ser. B (Methodol.) **58**(1), 267–288 (1996)
50. M. Unser, P.D. Tafti, Stochastic models for sparse and piecewise-smooth signals. IEEE Tans. Signal Process. **59**(3), 989–1006 (2011)
51. M. Unser, A. Aldroubi, M. Eden, B-spline signal processing. I. Theory. IEEE Trans. Signal Process. **41**(2), 821–832 (1993)
52. M. Unser, A. Aldroubi, M. Eden, B-spline signal processing. II. Efficient design and applications. IEEE Trans. Signal Process. **41**(2), 834–848 (1993)
53. A.P. Vinod, V. Sivaramakrishnan, M.M. Oishi, Piecewise-affine approximation-based stochastic optimal control with gaussian joint chance constraints, in *American Control Conference (ACC)* (IEEE, Philadelphia, 2019)
54. S. Vovk, V. Borulko, Determination of amplitude levels of the piecewise constant signal by using polynomial approximation. Radioelectron. Commun. Syst. **60**(3), 141–153 (2017)
55. S.J. Wright, R.D. Novak, M.A.T. Figueiredo, Sparse reconstruction by separable approximation. IEEE Trans. Signal Process. **57**(7), 2479–2493 (2009)
56. X. Zhou, J. Liu, X. Wan et al., Piecewise-constant and low-rank approximation for identification of recurrent copy number variations. Bioinformatics **30**(14), 1943–1949 (2014)

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